Investigation: IGB ELETRONICA S.A - EM RECUPERACAO JUDICIAL (CVM code 6815), fiscal year 2012, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Comunicação e Informática (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 2, filed on 2023-06-01): IGBR3 (Ações Ordinárias, Bolsa, since 1973-12-19).

Document used

2012 DFP, version 4, filed on 2013-11-27; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2013-03-28); 2 (2013-04-18); 3 (2013-04-24); 4 (2013-11-27).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=33276

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DSRI, GMI, SGI, DEPI, SGAI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: zero denominator: DSRI, t−1 revenue
- M-score component; it has no rule of its own
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

GMI, gross margin index
- Value: zero denominator: GMI, t−1 revenue
- M-score component; it has no rule of its own
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

AQI, asset quality index
- Value: 1.024 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 59 (n = 468)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: zero denominator: SGI, t−1 revenue
- M-score component; it has no rule of its own
- Percentile in sector and year: —

Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: zero denominator: SGAI, t−1 revenue
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

TATA, total accruals to total assets
Value: 0.263 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 99 (n = 487)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.130 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 76 (n = 480)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: zero denominator: DPO, t−1 cost of sales
Alert by the study 1 rule: does not trigger alerts
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 32.5 days)
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −4.83 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 6 (n = 439)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.253 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 98 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 0.689 (index)
Components: X1 = −0.052; X2 = −0.135; X3 = 0.230; X4 = −0.070 (indices)
Percentile in sector and year: 33 (n = 609); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (7 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): —
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 8 accounts above the R$1,000 tolerance (group a, totals and key lines: 1; group b, reclassifications: 7), in the annex at the end.

What was known on the date

- DFP 2010, v1: 2011-04-27
- DFP 2010, v2: 2011-04-27

- DFP 2011, v1: 2012-04-13
- FCA 2012, v1: 2012-04-13
- FCA 2012, v2: 2012-07-09
- FCA 2012, v3: 2012-09-14
- FCA 2012, v4: 2012-12-07
- DFP 2012, v1: 2013-03-28
- DFP 2012, v2: 2013-04-18
- DFP 2012, v3: 2013-04-24
- DFP 2011, v2: 2013-11-26
- DFP 2012, v4: 2013-11-27
- DFP 2013, v1: 2014-04-29

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
Excerpt checked literally: "Em nossa opinião, exceto pelos efeitos dos assuntos descritos no parágrafo Base para opinião com ressalva sobre as demonstrações financeiras individuais, essas demonstrações financeiras acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da IGB Eletrônica S.A. em 31 de dezembro de 2012, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2021 FCA: Grant Thornton Auditores Independentes Ltda.
Auditor in the 2022 FCA: Grant Thornton Auditores Independentes Ltda.
Auditor in the 2023 FCA: Grant Thornton Auditores Independentes Ltda.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2013: BC Control Auditores Independentes -> individual auditor (name omitted). The outgoing firm audited 1 consecutive fiscal years (2010 to 2010), by the start and end dates in the FCA.
- 2014: individual auditor (name omitted) -> Verdus Auditores independentes. The outgoing firm audited 2 consecutive fiscal years (2012 to 2013), by the start and end dates in the FCA.
- 2018: Verdus Auditores independentes -> Conatus Auditores Independentes S.S. The outgoing firm audited 5 consecutive fiscal years (2014 to 2018), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2020: Conatus Auditores Independentes S.S -> Grant Thornton Auditores Independentes Ltda. The outgoing firm audited 2 consecutive fiscal years (2019 to 2020), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC

(Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:38:21+00:00.

Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) DRE 3.01 Receita de Venda de Bens e/ou Serviços: 13.3 in the version used, 14.5 in the comparative (difference of 1.2)
(b) DRE 3.04.05 Outras Despesas Operacionais: 0.0 in the version used, −21.5 in the comparative (difference of −21.5)
(b) DRE 3.04 Despesas/Receitas Operacionais: 160.6 in the version used, 149.4 in the comparative (difference of −11.3)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: 171.0 in the version used, 159.7 in the comparative (difference of −11.3)
(b) DRE 3.06 Resultado Financeiro: 0.9 in the version used, 12.2 in the comparative (difference of 11.3)
(b) DRE 3.06.02 Despesas Financeiras: −18.8 in the version used, −7.5 in the comparative (difference of 11.3)
(b) DRE 3.04.04 Outras Receitas Operacionais: 179.7 in the version used, 189.9 in the comparative (difference of 10.2)
(b) DRE 3.02 Custo dos Bens e/ou Serviços Vendidos: −2.9 in the version used, −4.1 in the comparative (difference of −1.2)